

Git

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Git is a distributed version control system (DVCS) used to track changes in source and manage software development projects.

It was created by Linus Torvalds in 2005 for the development of the Linux kernel.

Simple Definition

Git is a tool that helps developers:

- Track changes in files
- Maintain different versions of a project
- Collaborate with other developers
- Restore previous versions when needed

Why do we need Git?

Imagine you are developing a project.

Without Git

```
Project
|
|-->Project_final.java
|-->Project_Final_New.java
|-->Project_Final_Latest.java
|-->Project_Final_Really_Final.java
```

Managing versions becomes difficult.

With Git

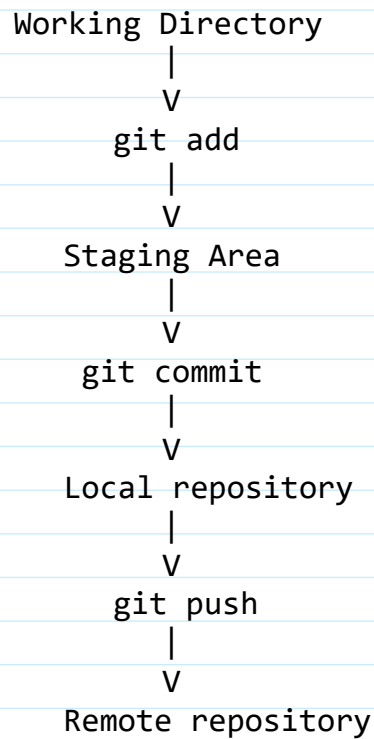
```
Project
|
|==>Git Repository
|
|==>version 1
|==>version 2
|==>version 3
|==>version 4
```

git automatically tracks every change.

Features of Git

1. **Version Control:** Tracks every modification made to files
2. **Distributed System:** Every developer has a complete copy of the repository.
3. **Fast Performance:** Operations are performed locally.
4. **Branching:** Allows development of new features without affecting the main project.
5. **Collaboration:** Multiple developers can work on the same project simultaneously.
6. **Backup and Recovery:** Previous versions can be restored easily.

Git Architecture



Git Terminology

Repository(Repo)

A repository is a storage location for your project.

Project Files + Git History = Repository

Commit: A commit is a snapshot of your project at a specific point in time.

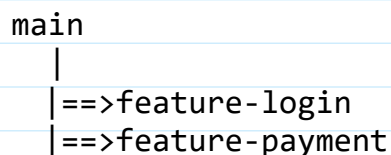
Example:

Commit 1: Initial Project

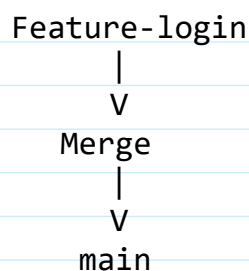
Commit 2: Added Login

Commit 3: Fixed Bugs

Branch: A branch is an independent line of development.



Merge :Combine changes from different branches.



To create a git repository we have to give:

```
git init
```

Output: Initialized empty git repository

.git is the hidden folder. This folder stores all Git information.